

Post-Flight Analysis of the Stardust Sample Return Capsule Earth Entry

Dean A. Kontinos*

Ames Research Center, Moffett Field, CA, 94035

and

Mairead Stackpoole†

ELORET Corp., Sunnyvale, CA 94086

This paper presents an overview of post-flight activities for assessing the entry performance of the Stardust Sample Return Capsule and the analysis tools used to design it. Three sources of information are leveraged: the recovered return capsule thermal protection system, airborne observations of the entry using instruments that provide spectral resolution of the hot return capsule and shock layer gasses, and radar signature during the terminal descent stage. The paper describes the objectives of the post-flight analysis, the information sources, and the methods of analysis. Sample results from detailed analyses are presented.

I. Introduction

In the early morning of January 15, 2006, the Stardust Sample Return Capsule (SRC) successfully delivered its precious cargo of cometary ejecta particles to the awaiting recovery team at the Utah Test and Training Range (UTTR). The SRC returned to Earth at 12.8 km/s (inertial): the fastest human-made object to traverse our atmosphere and only the second super-orbital velocity entry since the Apollo Program (the previous being the Genesis SRC). In addition to the excitement that comes with the rarity of obtaining extra-terrestrial material, it was a remarkable event in entry physics. At this entry speed, the atmospheric gas in the shock layer preceding and surrounding the SRC was dissociated and glowing hot, so hot that pre-flight analysis predicted that roughly 10% of the heat flux to the body at peak heating is due to radiant energy from the shock layer. In addition, the thermal protection system (TPS) was the first use of a NASA-developed lightweight material that manages the incident heat flux by means of ablation. In a way, the heatshield was designed to sacrificially boil away from the SRC thereby diverting the heat energy away from its innards. The ablation products were mixed into the shock layer thereby creating a complex, and remarkable, fluid-material interaction at very high energy.

In the intervening time between the launch and return of Stardust, NASA began the Exploration Program with a goal of returning humans to the moon. An element of that architecture is a Crew Exploration Vehicle (CEV or so-called Orion spacecraft) that is to reenter the earth's atmosphere at super-orbital velocity from a trans-lunar trajectory. The CEV shape is a derivative of Apollo, and, hence, is in the class of blunt-body capsules as is the Stardust SRC. Furthermore, the predicted aerothermal environment on the CEV during lunar return is on the same order as that sustained by the Stardust SRC. Hence there is keen interest by the Exploration Program on the flight performance of the Stardust SRC entry system: aerodynamic, aerothermodynamic, and thermal protection. Of utmost importance is an evaluation of the Phenolic Impregnated Carbon Ablator¹ (PICA) forebody heatshield material as that is a leading choice for a lunar-return-capable CEV.

This paper presents an overview of post-flight activities for assessing the performance of the SRC and the analysis tools used to design it. Three sources of information are leveraged: the recovered SRC thermal protection system, airborne observation of the entry using instruments that provide spectral resolution of the hot SRC and shock layer gasses, and radar signature during the terminal descent stage. The objectives of the post-flight analysis, the information sources, and the methods of analysis are described. Sample results from in-depth analyses are presented.

* Fundamental Aeronautics Program Technical Integration Manager, Associate Fellow AIAA.

† Research Scientist.

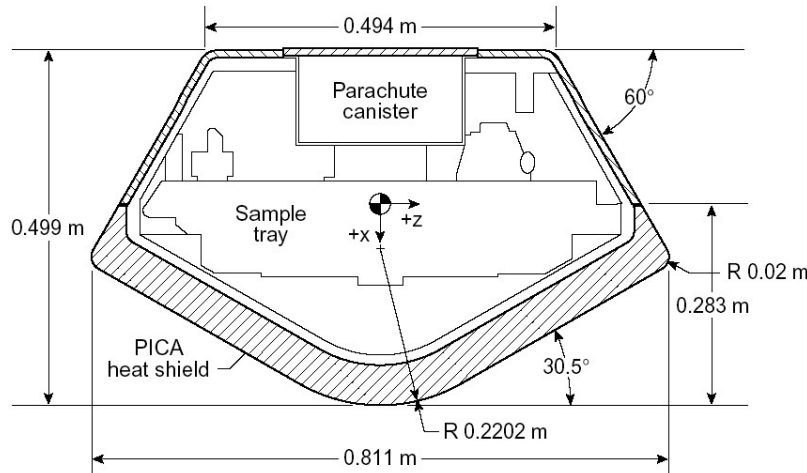


Figure 1. Stardust Sample Return Capsule.

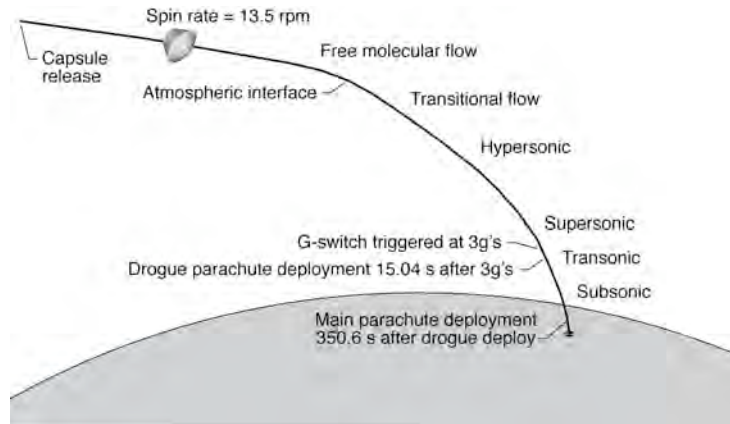


Figure 2. Nominal Stardust Entry, Descent and Landing Sequence

II. SRC Description

The Stardust SRC was built by the Lockheed Martin Space Systems Company. The aeroshell was comprised of two primary components:² a 60 degree sphere-cone forebody heatshield and a 30 degree truncated cone backshell (see Fig. 1). The width at maximum diameter was 0.827 m. The flight path was nominally purely ballistic (non-lifting trajectory) with a ballistic coefficient of $\sim 60 \text{ kg/m}^2$. The expected upper bound of the heating environment was a peak heating of $\sim 1200 \text{ W/cm}^2$ at the forebody stagnation point and a heat load of $\sim 36 \text{ kJ/cm}^2$. Of the incident heat flux, about 10% was expected to be from shock layer radiation at the time of peak heating.

For comparison, the expected heating environment for CEV is a peak heating of $\sim 900 \text{ W/cm}^2$ and 70 kJ/cm^2 . One major difference between the Stardust SRC and CEV is that CEV is to fly a lifting trajectory (lift generated by maintaining the vehicle at an angle-of-attack through offset of the center of gravity) and therefore the peak heating location is neither at the geometric center or stagnation point. Furthermore, the fractional amount of heating due to shock layer radiation is predicted to be significantly higher for CEV than for Stardust. At the time of peak heating, $\sim 40\%$ of the incident heat flux is expected to be due to shock layer radiation. Although the relative entry velocity of CEV upon return from the moon will be lower than Stardust, 10.8 km/s for the former and 12.6 km/s for the latter, the CEV will be substantially bigger: 5 m at maximum diameter. The increased size will create a larger amount of high temperature gas in the shock layer and therefore increase the radiant source. Differences aside, there is much commonality between the two environments from a fundamental physics aspect.

The main thermal component of the Stardust forebody heatshield was PICA¹ material, developed at NASA Ames Research Center under the lightweight ceramic ablator development program in the '80s. It consisted of a rigid carbon fibrous substrate infiltrated with phenolic resin yielding a TPS with good ablation and pyrolysis behavior. In addition, PICA had the advantages of low density ($\sim 0.27\text{g/cc}$) coupled with efficient ablative capability at high heat fluxes making it an enabling technology for the Stardust mission. In comparison to conventional high-density carbon phenolic TPS solutions, PICA's thermal conductivity was substantially lower.

The backshell TPS was Super Lightweight Ablator³ (SLA)-561V - a Lockheed Martin ablative TPS developed in the '60s. It has substantial flight heritage: it was used on the forebody heatshields of all NASA missions to Mars (e.g. Viking, Pathfinder) and on portions of the Shuttle external tank. SLA-561V consisted of an ablative component packed in a honeycomb core. The core was pre-bonded to the substructure allowing construction of a large component.

The entry, descent and landing of the SRC occurred as depicted in Fig. 2. Upon release from the spacecraft bus, the SRC was spun to approximately 16 rpm. The spinning aeroshell provided unguided aerodynamic stability and thermal protection as it decelerated to low supersonic Mach numbers.^{4,5} Near Mach 1.4 and $\sim 32\text{ km}$, a drogue chute was deployed to provide dynamic stability as the baseline SRC became increasingly unstable approaching transonic speeds. At Mach 0.16 and 3.3 km the descent parachute was deployed.⁶ The SRC descended to the ground under chute. The design specification was a maximum impact velocity of 4.6 m/s . The authors are not knowledgeable of the actual impact speed or force.

III. Analysis Objectives

The Orion Thermal Protection System Advanced Development Project, the Jet Propulsion Laboratory (JPL) Stardust Project and the NASA Engineering and Safety Center (NESC) are jointly performing an integrated reconstruction of the SRC atmospheric entry to assess the performance of the entry system and, importantly for the CEV design, the accuracy of the aerodynamic, aerothermodynamic, and TPS response tools and processes used to design the SRC; these very same tools are being used in CEV design.

The objectives of the integrated analysis are to 1) assess aerodynamic performance of the SRC as compared to design prediction, 2) confirm conformance to anticipated trajectory dispersions, 3) assess TPS surface temperature and ablation response as compared to the pre-flight model, 4) assess convective and radiative aerothermodynamic heating models, 5) compare characteristics of flown TPS to arc-jet tested samples to confirm the verification and validation process, and 6) provide guidance for Orion flight test objectives and instrumentation approach. Analysis of the recovered hardware is essential to meeting the broader goals stated above. Specific objectives of the hardware analysis are the following:

Forebody Heatshield

- 1) Determine unusual surface features indicating off-nominal aerodynamic performance, off-nominal TPS performance, or pre-entry damage.
- 2) Measure in-depth char and transition layer of TPS at select locations to determine spatially varying integrated heat load.
- 3) Measure in-depth properties of TPS, e.g. density, chemical composition, and thermal properties, to compare to pre-flight models and arc-jet tested samples.
- 4) Measure residual bond strength to assess aging effects

Backshell

- 1) Determine unusual surface features indicating off-nominal aerodynamic performance, off-nominal TPS performance, or pre-entry damage.
- 2) Measure surface reflectivity as indicator of surface heating, flow pattern, or ablation product deposition.
- 3) Measure in-depth char and transition layer of TPS at select locations to determine spatially varying integrated heat load.

To date, the post-flight analysis team has concentrated mostly on the forebody heatshield.

IV. Data Sources

Unfortunately, the SRC was not instrumented and, therefore, there are no time-resolved direct measurements of the state of the aeroshell, e.g. acceleration, temperature, pressure, etc. However, an auxiliary mission to observe the

entry from an airborne platform was successfully executed. These data, in combination with pre-flight specifications, in-space navigation, terminal descent radar, and recovered hardware, constitute a rich source of evidence with which to assess the performance of the entry system.

A. Airborne Observation

The flight path from the Pacific ocean across to UTTR provided viewers in California, Nevada, and Utah a visually stunning meteor sighting - a meteor whose timing and flight path were quite predictable. Leveraging the methodologies and personnel from the meteor observation community, a team of researchers imaged the SRC entry aboard the NASA DC-8 airborne observatory.⁷ At SRC entry, the airplane was at 39 kft positioned within 4 miles of the prescribed, preferred target view location in Nevada just outside the western boundary of UTTR. The incoming SRC was first acquired approximately 18 s after atmospheric interface and tracked for approximately 60 s, an observation period that was roughly centered in time around predicted peak heating. The radiative signal from the SRC and surrounding shock layer gases were measured by 15 of 18 instruments that had various combinations of spectral range, spectral resolution, and temporal resolution (note that there was no spatial resolution of the SRC; it appeared to all the cameras as a point source). The data were assessed to be of good quality and sufficient to address all observation objectives: absolute radiance, spectral resolution of shock layer emission, and wake train evolution. Further details of the observation campaign flight, instrument suite, and acquired data can be found in Ref. 7. In addition to the instruments aboard the aircraft, there were complementary ground observations from sites along the track.

B. Terminal Descent Radar

The incoming SRC was acquired prior to drogue chute deployment by ground radar stationed at UTTR at approximately Mach 2 and 35 km. The SRC was tracked through drogue chute deployment, main chute deployment, and descent under canopy – a total duration of ~11.5 minutes. From range and timing data, the vehicle velocity and absolute position is known through that time period.

C. Recovered Hardware

As a consequence of design requirements for safe and intact return of the sample collection array, the SRC landing was a fairly benign event to the aeroshell. Based on imprints in the dirt and marks on the aeroshell, it is surmised⁸ that the SRC first impacted on the forebody aeroshell nose approximately 15° from the symmetry axis. It then bounced a total of 4 times. After the last small bounce, the SRC ended up on its rim, at which time it rolled on its edge in a weaving manner downwind for a total of about ~25 m. Near the end of the roll, the SRC had slowed enough that several times it flopped down on the side of the backshell briefly before coming back up to roll on the rim a little further. Finally the SRC tipped onto the side of its backshell for good, rolled in almost a complete circle, and came to rest on its side. Figure 3 shows the SRC in its final resting position. The initial impact location near the nose and the edge bounce location are evident. Also seen in the photo are parachute bridles extending from the rear. The parachute was released at the riser by a cutter triggered by the deceleration at impact. The parachute canister lid that forms the rear enclosure of the backshell would have come down at another location and was not found.

The aeroshell was lifted whole and transported to a processing facility at UTTR.⁹ In a Class 10000 clean room, the SRC was separated into its three primary structural pieces: the forebody heatshield, backshell, and sample return canister. The forebody heatshield component extended beyond the maximum diameter of the SRC and comprises about one quarter the length of the aft-facing flank. The sample canister was attached to the avionics deck within the forebody aeroshell component. The sample canister was separated from the heatshield and stored in a special GN₂ purged container. The heatshield and backshell were placed in individual, unpurged containers. All hardware – including the three major components of sample canister, heatshield, and backshell – were then transported to Johnson Space Center (JSC) on 17 Jan 2006. At JSC, the sample canister was moved to the class 100 Stardust cleanroom and opened. Sample recovery and analysis activities were then begun by the Stardust Science Team. The other components (heatshield and backshell) were curated as space exposed hardware in the Space Exposed Hardware (SEH) cleanroom.

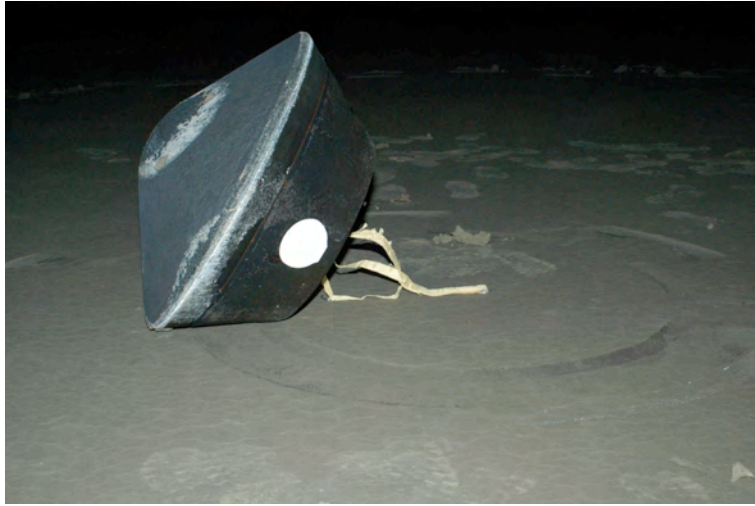


Figure 3. Photograph of SRC after landing and prior to recovery (courtesy Scott Sandford).

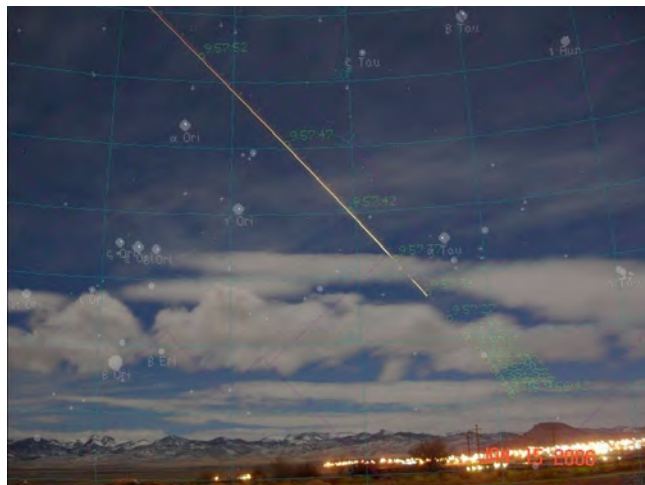


Figure 4. Photograph of SRC transit from Wendover Airfield, UT with star map overlay (courtesy Bruce Fischer).

V. Analysis

Several analysis activities have been initiated towards achieving the objectives listed previously and leveraging the aforementioned data sources,. These activities are described herein along with a description of results. Detailed reports describing the various analyses are referenced throughout.

A. Trajectory

The essential first step in the post-flight analysis is an estimate of the as-flown trajectory. The trajectory information provides flight conditions (altitude, velocity, and orientation) from which the aero/thermal environment is then assessed. Key data sources are the in-space navigation, airborne observation, ground photos, and terminal descent radar. As described in Ref. 10, the program POST¹¹ is used to propagate trajectories from the entry interface condition through the beginning of radar acquisition to the point of drogue chute deployment. Nominal atmospheric or aerodynamic properties are adjusted to reproduce the known trajectory state conditions.

In addition to the above, there are long-duration exposures of the SRC as it streaked across the sky. Using the starfield as reference and knowing the location of the viewpoint, two ground images are used to determine the flight

path.¹² One such image is shown in Fig. 4. This method yields only the SRC track, not the timing, i.e. one knows where it flew but not when. The timing is determined using imagery from the airborne observation. These temporally-resolved, absolute-time-stamped images (recorded on video tape) show the SRC moving against the starfield. Since the airplane location is known by on-board GPS, the timing of the SRC trajectory is determined by the intersection of the aircraft-SRC line of sight and the photographically determined flight path. These data provide the only trajectory information during the hypersonic flight regime.

Using the above analysis methods, the SRC is determined to have flown a nominal trajectory. Said another way, within the uncertainty of the analysis methods, there is no evidence that the SRC flew anything but a nominal flight profile as predicted prior to entry. Therefore, all subsequent analyses employ this nominal flight trajectory. The fact that the entry trajectory was nominal is evidence that the SRC aerodynamic model and trajectory simulation are accurate.

B. Recovered Hardware

The heatshield and backshell are the two components of interest for this analysis. The heatshield is shown in Fig. 5. It is in remarkable condition with few obvious surface features caused by the landing. In the photo, the bounce locations on the nose and edge are evident. Also, seen are rub locations on the edge from when it rolled. The backshell is shown in Fig. 6. The holes in the flank region are extracted core samples that are described later. The backshell is also in remarkable conditions without any obvious impact related surface features.

The shape of the two components is determined¹³ using a newly developed LASER scanning system¹⁴ designed to assess damage to Space Shuttle tiles. The original handheld system is modified to accommodate the larger - relative to Shuttle tile - SRC heatshield and backshell. Components of the instrument include a digital camera, a LASER diode line generator, and an electronics subsystem that coordinates image acquisition, processing, motion control and communications. The 3D images are generated through LASER triangulation.

The entirety of the outer surfaces are scanned to a resolution of ~1 mm. Figure 7 shows results of the scan. Lost TPS material from impact and accumulation of ground material are clearly evident. These data are used to assess the changes in configuration from as-built conditions. Presumably, the primary causes of surface change are the obvious damage caused from impact and ablative recession. A major difficulty in utilizing the geometric data is the virtual mating of the two pieces and referencing to an absolute datum. Since it is theoretically possible, and for the forebody completely true, that the entirety of the surface is recessed from its pre-flight condition, it is not obvious how to reference the existing coordinates with respect to the pre-flight geometry. The referencing is accomplished by first clocking the backshell based on non-symmetric surface features (bolt locations and rear-face attachment points). The remaining coordinate is fixed by reference to the metallic features on the rear-face (push-off plates, parachute canister, etc.) that are assumed to be their original shape. The forebody is clocked to the backshell by the specific attachment locations in the interior (not scanned). The primary uncertainty is in the virtual mating of the two pieces and the amount of play at the plane of the forebody and backshell interface. Keep in mind that the underlying data are individual points and not geometric objects such as lines and planes.

Using the LASER scan data, eyewitness examination of the hardware, and photographs, locations for sample extractions were identified by a team of stakeholders from NASA and Lockheed Martin Space Systems Company (LMSS). On the forebody two primary cores, a provisional core, and an edge slice were selected for sample extraction as shown in Fig. 8. Core 1 is near the apex. Since the SRC impacted off-center from the nose, a location off-center and away from the primary impact point, yet still on the spherical segment, is chosen. Core 2 is at the mid-point of the flank region. The provisional core is the mirror location of Core 2. Finally, a segment at the edge along the ray of Core 2 is selected. Cores are 1.75 inches in diameter.

On the backshell, four core locations are identified. The first is on the mid-point of the flank. The second is at a location where the SLA had an apparent stitch line of its honeycomb structure. The third encapsulates a ~3/4 inch diameter repair plug, a consequence of a launch integration umbilical connection. Finally, the fourth core is on the rear. Fig. 6 shows the backshell with the first 3 cores extracted. All forebody and aftshell cores described above, except the provisional flank core, have been extracted by LMSS personnel under supervision by JSC curation. The LASER scanned geometric data are recalibrated so that recession at the core locations match recession deduced by measuring residual material up from the bondline.

C. Heating and TPS Response

Using the trajectory described above, a coupled aerothermodynamic and material response analysis is performed.¹⁵ Heating to the SRC is estimated using the computational fluid dynamic (CFD) code DPLR,¹⁶ which is a parallel, multiblock, finite-volume code that solves the reacting Navier-Stokes equations including finite rate chemistry and thermal nonequilibrium. The flow is assumed to be in thermal nonequilibrium and is modeled with a



Figure 5. Photograph of recovered SRC forebody heatshield.



Figure 6. Photograph of recovered SRC backshell after extraction of flank cores.

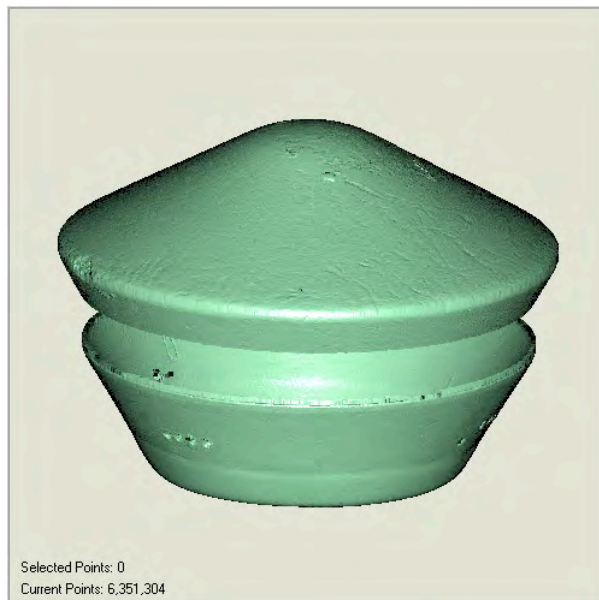


Figure 7. LASER scanned heatshield and backshell virtual objects.

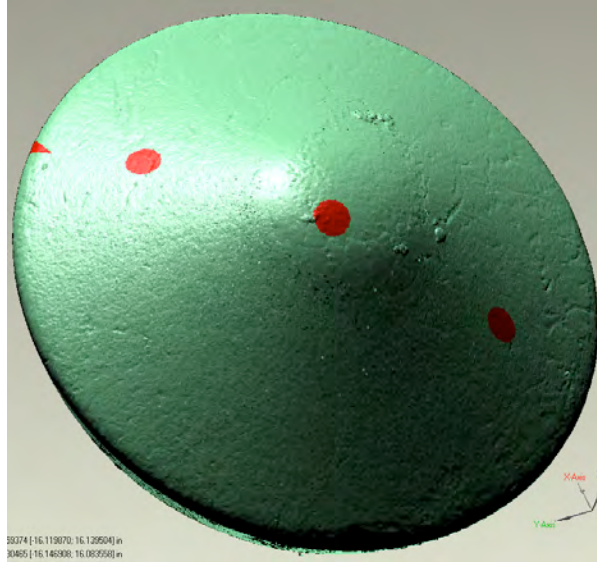


Figure 8. LASER scanned heatshield virtual object with red marks denoting specified sample locations.

two-temperature, 11 species and 19 reactions, finite-rate air chemistry model. The wall catalysis is modeled with constant catalytic efficiency and is fully catalytic to ions. The flowfield is assumed axisymmetric about the SRC at zero angle-of-attack. The complete fore- and aft-body solution is computed. The flow is assumed to be laminar throughout the trajectory.¹⁷ The radiative heating component is neglected in this analysis. Auxiliary computations show the radiative component to be 9% of the total peak heat rate and 4% of the heat load at the stagnation point. At other points on the surface the fractional contributions to heat rate and load from radiative heating are less.

To define the heat pulse, 21 CFD solutions are computed along the trajectory, the earliest being at ~81 km altitude and ~12.4 km/s velocity, and the latest being ~40 km altitude and ~1.3 km/s velocity. Based on Knudsen number, the flow at the earliest CFD time is on the border of being non-continuum. The heat pulse is completed by interpolating the CFD-produced surface heat flux between solutions. At times earlier and later than the period characterized by CFD, the local surface quantities are scaled by freestream conditions to close out the heating profile. The convective heating profiles at the stagnation point and two core locations are shown in Fig. 9.

The TPS material response is computed using the ablative thermal analysis code FIAT v2.4,¹⁸ which models the surface energy balance in conjunction with ablative decomposition of the PICA material and in-depth conduction. The material stack consists of PICA (5.8 cm), HT-424 (0.14 cm) and aluminum 2024 (1.27cm). All materials have an assumed initial temperature of -20°C. There is no contact resistance at any material interfaces and the back wall is assumed adiabatic. The model is one-dimensional in-depth. The calculations are run to 1000s to properly account for the conduction and ultimate rejection of the heat absorbed during entry. The FIAT computations yield the in-depth, temporally-resolved temperature profile within the stack and the ablative response. The ablative response is comprised of the in-depth pyrolysis gas production, pyrolysis gas convection to the surface (surface blowing), and surface recession. The input heat pulse is calculated without regard to the surface blowing or recession.

The CFD (DPLR) and material response (FIAT) codes are loosely coupled to enforce consistency of surface temperature. Initially, CFD solutions are calculated assuming a surface in radiative equilibrium, i.e. there is no internal conduction into the surface and the wall temperature is that that produces a convective heat flux equal to the surface re-radiation. The resultant heating profile is input to FIAT and a new surface temperature history is produced that accounts for the effects of material response. A FIAT solution is calculated at each point on the surface CFD grid. The FIAT-generated surface temperatures are imposed as boundary conditions within the CFD and the heat pulse is updated. In practice, the process converges in one iteration.

Three groups of comparisons are possible between the combined trajectory / aerothermal-environment / material-response simulation and available flight data. One group is the character of the recovered TPS compared to the state calculated at the end of the time period; key quantities are the in-depth density profile and the surface recession at the core locations. These comparisons are shown in the Material Characterization section. The other two groups exploit the data from the airborne observation. Since there was no on-board instrumentation, the only temporally-resolved data available to inform the entry performance is that from the airborne observation. Recall that

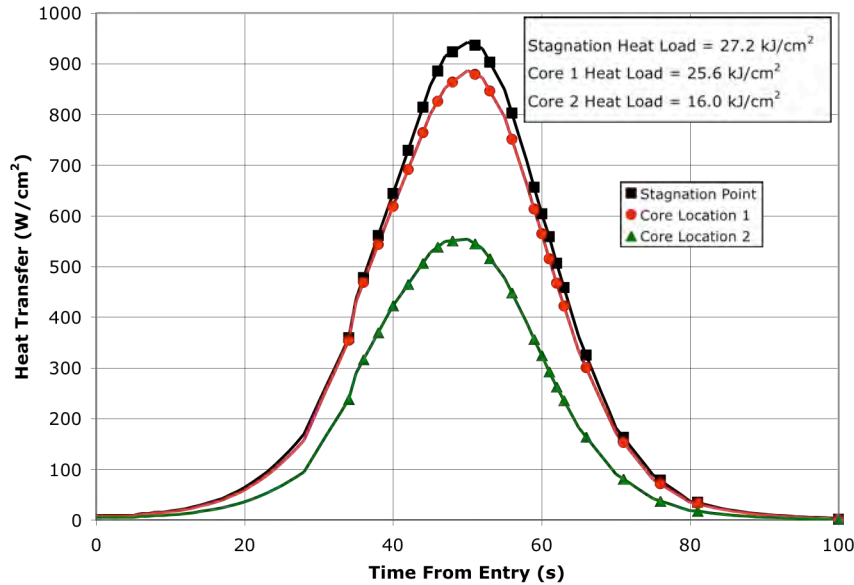


Figure 9. Estimated SRC heat flux histories at the stagnation point, Core 1 location, and Core 2 location; symbols indicate times when CFD solutions are computed.

these data are temporally and spectrally resolved; they are not spatially resolved. As shown in Ref. 7, the recorded signals are emission from the air plasma at specific wavelengths superimposed on broadband emission from the hot heatshield. The two light sources can be effectively analyzed separately. Analysis of the air emission is presented in the Flowfield Radiation section. Attention is now focused on the broadband emission.

The primary data are spectra obtained using an Echelle spectrograph during the period 34-44s from entry.¹⁹ This period spans the increase in heating during the front half of the heat pulse (see Fig. 9). Since the emission is not spatially resolved, a given spectra is a measure of the entirety of the light emitted from the shock layer and glowing hot heatshield (corrected for atmospheric absorption and scattering) at the moment in time. The Echelle camera captured light in the wavelength range 368-872 nm at 0.14-0.19 nm resolution. The simulation data are processed to match in kind the observation data. Synthetic broadband spectra are generated by integrating grey body emission over the surface of the heatshield, accounting for local temperature conditions and view angle to the SRC. This synthetic signature is compared to flight data at 44s from entry in Fig. 10 from Ref. 15. There is remarkably good comparison between calculation and flight data at this time. Also shown are synthetic curves generated by assuming ± 50 K offsets in surface temperature. The curves are sufficiently sensitive to surface temperature to permit a judgment that the comparison is accurate to within 50 K. Keep in mind that these comparisons are a measure of average surface temperature. It is not possible to directly determine surface temperature distribution from the observation data. As shown in Ref. 15, the simulation compares well to the Echelle data during the period 38-44 s from entry. Prior to 38 s, the simulation over-predicts the broadband emission, and by extension the surface temperature. The cause of this discrepancy is not known. One possibility is that the simulation does not model the ablation of white thermal control paint that was on the SRC. There is spectral evidence that the paint was still ablating during the time 34-38 s from entry.¹⁹ How this paint layer might affect the thermal response and emission is not known.

D. Flowfield Radiation

As mentioned previously, the data from the airborne observation provide a temporally and spectrally resolved record of the SRC entry. The signal is a superposition of the air plasma emission from the gasses in the hot shock layer and the broadband emission from the hot heatshield. This section describes the analysis of the air plasma emission. It is this emission that radiatively heats the vehicle during entry.

The emission is calculated²⁰ by post-processing the DPLR solutions using the radiation production and transport code NEQAIR,²¹ which solves the radiative specific intensity by integrating the Radiative Transfer Equation along a line-of-sight. In calculating the radiative heat flux, NEQAIR assumes the flow to vary one-dimensionally in the direction normal to the body and employs the tangent slab approximation for the angular integration. The frequency

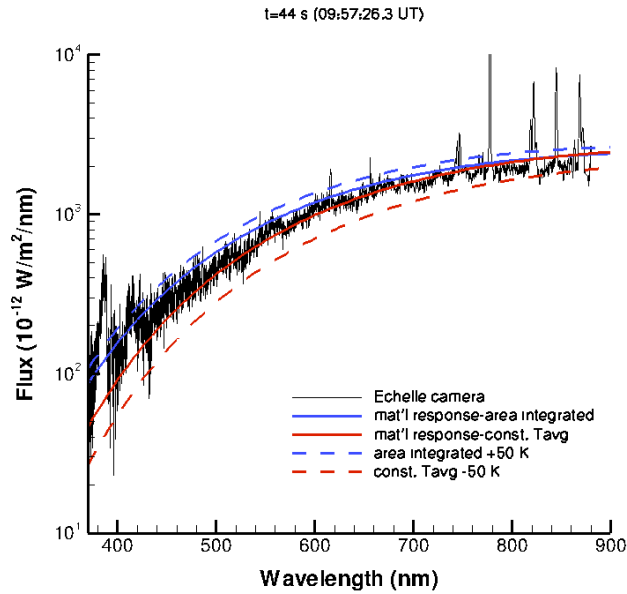


Figure 10. Synthetic broadband spectra from integrated simulation compared to Echelle spectrum at 44 s from entry (from Ref. 15).

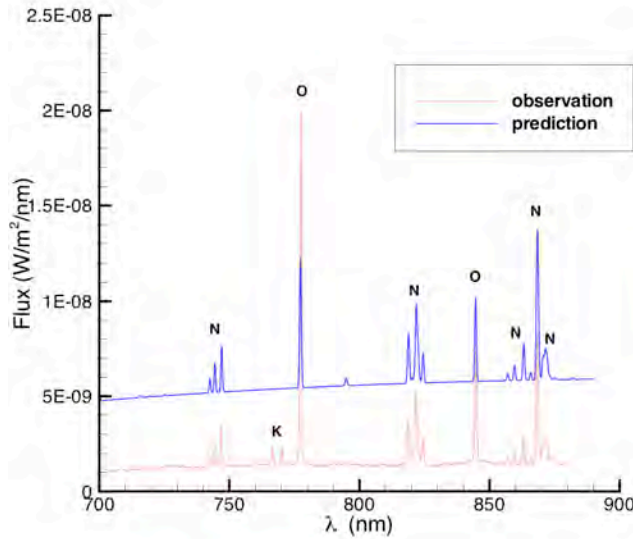


Figure 11. Synthetic line spectra compared to Echelle spectrum at 42 s from entry (from Ref. 20).

integration, however, is treated in detail line-by-line spectroscopically. NEQAIR further assumes that the electron impact process governs the radiation transition, and those rates are determined by a Quasi-Steady State model.

Traditionally, NEQAIR is used to estimate the radiative heating component to the surface of the vehicle. In this approach, lines-of-sight are generated normal to the surface. Using the tangent-slab approximation, the radiative heating is estimated from the column of gas as predicted by the CFD along the line of sight. This approach is usually sufficient for design and analysis of on-board spectral instrumentation. However, the airborne observation is a different situation. Here, the instruments are collecting all light from the entirety of the SRC and flowfield at a considerable distance along the line-of-sight from the airplane. Therefore a different analysis approach is developed.

The radiative production and transport is estimated along many lines-of-sight parallel to the view angle from the airplane. First, the axisymmetric DPLR solution is replicated circumferentially to create a three-dimensional

flowfield, which is required because the lines-of-sight are not exclusively in the symmetry plane. Then, NEQAIR is modified to reverse the integration direction, i.e. the radiative transport is computed from body, through the shock layer, past the shock wave itself. Furthermore, not all lines-of-sight intersect the body surface. In these cases, procedures are developed for determining end points of the integration.

After emission is calculated along individual lines-of-sight, the total is integrated to provide the gross emission from the SRC and shock layer. The signal is convolved with the instrument function to parse and bin the spectral energy similar to the record. Figure 11 from Ref. 20 shows data from the Echelle spectrograph along with a synthetic spectrum at 42 s from entry interface. Note that the continuum predictions (the underlying broadband signal) are several times higher than the observation. Since this radiation calculation is made using the DPLR solution with a radiative equilibrium surface assumption, the simulated surface temperature is several hundred degrees higher than actual. This difference in surface temperature causes the difference in the continuum emission. Because the air emission lines are at least an order of magnitude greater than the continuum, this difference in surface temperature does not significantly affect the comparison of peak emission. As shown in Fig. 11 line intensity (height above the continuum) matches well to the observation for atomic nitrogen lines. However, the prediction appears to underestimate the line intensity for atomic oxygen lines. Efforts are underway to understand this effect.

E. Material Characterization

The extracted cores are analyzed to characterize the end-state of the forebody heat shield material.²² Figure 12 shows photographs of the cores. The surface is fully charred and black in color. Clearly evident is color variation through the depth corresponding to the transition from fully-charred to virgin PICA material toward the bondline side where substantial virgin PICA is still present. The cylindrical cores are sub-divided with a semi-cylinder separated for destructive evaluation (the edge piece remains intact). The semi-cylinder is sectioned in-depth. Several laboratory tests are performed on the sectioned pieces to characterize the material as a function of depth from the char surface. Good correlation is found between, emissivity, material composition, resin content, and density going through the depth of the material. By these measures, the material survived the entry without anomaly.

A crucial comparison is the measured end-state of the material to that resulting from the analysis described in Heating and TPS Response section. The parameters of interest are the recession and in-depth density profiles. The recession is compared in Table 1 from Ref. 22. Note that at the stagnation point and core locations, the radiative heating component is estimated for inclusion in the recession calculation. Both regions show the simulation to over-predict the recession. The flank core recession is over-predicted by 51-61%. At this time, the reason for this discrepancy is not clear. From a design perspective, the simulation tools and procedure at least show conservatism with respect to recession. The calculated in-depth density profiles at the two core locations are compared to the measurements in Fig. 13. The data are referenced to their respective recessed surfaces, i.e. the differences in recession between measurement and computation are not captured in this plot. The computed in-depth density profiles compare extremely well to the measurements. The measurements and computations show approximately 6-7 mm of fully-charred material near the surface. There is a char transition zone to approximately 10 mm in-depth that is captured well by the material response model. Notice in the data and the computations that the char depth of the flank core is greater than that of the near-stagnation core. This result may at first seem counter-intuitive since the heat load is greater toward the stagnation point and hence one might anticipate a greater char depth in that region. However, the peak instantaneous heat rate is also greater at the stagnation point. The PICA ablator accommodates the increased heat rate (as compared to the flank region) by char removal through ablation. Thus for the case of Stardust, the near-stagnation core shows a greater recession but less char depth than the flank core. The combined trajectory-aerothermal-material-response simulation captures these trends. There is an over-estimate of the recession, but the density profiles are reproduced remarkably well.

VI. Future Work

There is an incredible wealth of information to be gleaned from the Stardust SRC and the airborne observation data; the activities described herein are those forebody heatshield analyses that are tractable using available tools. Analyses of the backshell are yet to have begun in earnest. On the heatshield, future work will require an increase in the complexity of the analysis methods. Several tasks are under consideration. First is a fully-coupled analysis between CFD and material response, wherein mass blowing information from the material response code is passed to the CFD for input as a surface boundary condition. The CFD code would contain species conservation equations and reaction models for the ablation products. The formation of CN would be modeled. Subsequently, the radiation production-transport code would simulate the emission from CN that could be compared to the airborne observation

Table 1: Measured and Calculated Recession (from Ref. 22)

		convective heating only		convective and radiative heating	
Location	Measured (cm)	Calculated (cm)	% difference from measured	Calculated (cm)	% difference from measured
Stagnation Point	-	0.96	-	1.04	-
Core 1	0.57±0.03	0.86	51	0.92	61
Core 2	0.32±0.02	0.39	22	0.4	25

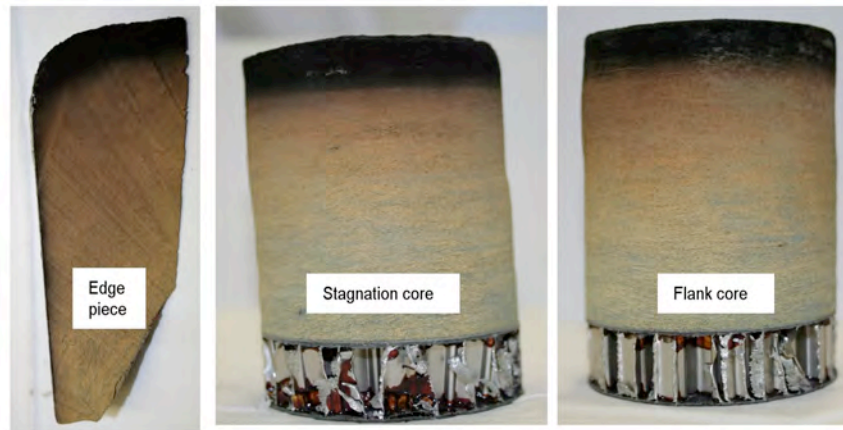


Figure 12. Photographs of cores extracted from SRC heatshield (from Ref. 22).

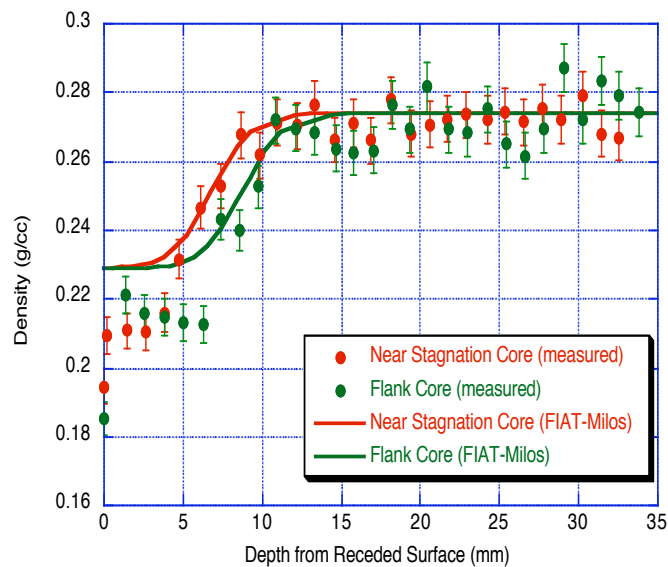


Figure 13. Comparison of measured and calculated density profiles for the near stagnation and flank cores. The x-axis is referenced to each receded surface (from Ref. 22).

data. Favorable comparisons would increase confidence in the integrated simulation capability. Also captured by the airborne observation are characteristic spectrum from K and Zn believed to be thermal control paint ablation products (from the forebody heatshield). These spectra are detected starting early in the entry trajectory. Unfortunately, the authors are not aware of an ablative model for the paint, but if such a model could be developed, then these spectra could be related to ablation rate and ultimately to heating rate in the early non-continuum flow regime of the trajectory. Finally, the wedge-shaped piece extracted from the edge of the heatshield shows a vivid char profile. This sample provides an opportunity for assessing multi-dimensional material response, and uncovering deficiencies in traditional one-dimensional material response models.

VII. Concluding Remarks

The recovered Stardust SRC hardware (with minimal impact damage) and airborne spectroscopic observation comprise a rare data set. The activities described herein make use of the data to provide a unique benchmark of entry system performance and analysis capability. Based on comparisons to date, the PICA material performed without anomaly, providing thermal protection and graceful recession. The nominal flight trajectory affirms the accuracy of the aerodynamic database and flight prediction. The integrated analysis process of trajectory, aerothermal environment, and material response show good comparison to flight data: most notably agreement in interpreted average surface temperature near peak heating, agreement in in-depth PICA density profiles, and reproduction of air plasma line emission. There are discrepancies to resolve. The recession is over-predicted by half near the stagnation point, less so at the mid-flank region. Also, the interpreted average surface temperature is over-predicted during the early portion of the trajectory. In the shock layer, the emission at lines characteristic of oxygen species are under-predicted. It is hoped that these discrepancies can be resolved through further analysis and comparison to additional data as they become available from other instruments flown in the observation campaign.

Serendipitously, the Stardust SRC entry serves as an early flight test of the PICA TPS material that is the primary candidate in consideration for the CEV. In fact, one would be hard-pressed to devise a more relevant flight profile as an analogue of CEV lunar return. Of course a real flight test would have had on-board instrumentation, however this deficiency is somewhat compensated by the airborne observation. In the near term, these analyses affirm the tools and processes being used to design the CEV TPS. In the mid term, the post-flight analysis experience of Stardust informs potential future flight tests of CEV with TPS related objectives. In the long term, these results increase the body of knowledge of entry performance for design and operation of future sample return and planetary entry vehicles.

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Contributors: Chris Dobell, David Hash, Joseph Lavelle, Creon Levit, Frank Milos, Yen Liu, George Raiche, Stefan Schuet, Kerry Trumble, Jeff Version, Michael Wright - *NASA Ames Research Center*; Betsy Pugel - *NASA Goddard Space Flight Center*; Karen McNamara - *NASA Johnson Space Center*; Prasun Desai, Garry Qualls - *NASA Langley Research Center*; Miria Finckenor - *NASA Marshall Space Flight Center*; Ioana Cozmuta, Dinesh Prabhu, Jerry Ridge, David Saunders, Steve Sepka, Matthew Switzer - *ELORET Corp.*; Peter Jenniskens - *SETI Institute*; Deborah Levin - *Pennsylvania State University*; Maegan Spencer - *Stanford University*; Iain Boyd - *University of Michigan*.

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